Week 1 Exercises

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May 9, 2022

Please complete all exercises below WITHOUT using any libraries/packages.

Exercise 1

Assign 10 to the variable x. Assign 5 to the variable y. Assign 20 to the variable z.

```
x <- 10
y <- 5
z <- 20
```

Exercise 2

Show that x is less than z but greater than y.

Note: your output must be a SINGLE boolean, do not output a boolean for each expression.

```
(x < z) & (x > y)
```

```
## [1] TRUE
```

Exercise 3

Show that x and y do not equal z.

Note: your output must be a SINGLE boolean, do not output a boolean for each expression.

```
(x != z) & (y != z)
```

```
## [1] TRUE
```

Exercise 4

Show that the formula `x + 2y = z`.

Note: your output must be a SINGLE boolean

```
(x + 2*y )== z
```

```
## [1] TRUE
```

Exercise 5

I have created a vector (test_vector) of integers for you. Determine if any of x, y, or z are in the vector.

Note: your output must be a SINGLE boolean, do not output a boolean for each expression.

```
test_vector <- c(1,5,11:22)


any(c(x, y, z) %in% test_vector)
```

```
## [1] TRUE
```

Exercise 6

Show which value is contained in the test vector. To do this you will need to create an element-wise logical vector using operators. `x == vector`. Once you have done that you will need to use slicing to return all indices that have matches. **Note: your output should be two integers**

```
test_vector <- c(1, 5, 11:22)
f <- 1
g <- 20


comparison <- (f == test_vector) | (g == test_vector)

match_test <- which(comparison)


print(match_test)
```

```
## [1]  1 12
```

```
#output is 1 and 12, the first and twelfth indices in our test_vector which is 1 and 20.
```